

From Rainfall to Risk : Predicting Flood Hazards



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1.INTRODUCTION

1.1 Problem Statement

Flooding is one of the most economically and socially devastating natural disasters worldwide, and Sri Lanka is no exception: monsoonal rainfall, river overflow, and rapid urbanisation that erodes natural drainage capacity combine to put large populations at recurring risk. Accurate, location-specific flood risk prediction enables governments, emergency services, urban planners and insurers to act preventively - prioritising evacuation planning, infrastructure investment and resource allocation toward the areas that need it most.

This project frames flood risk prediction as a continuous regression problem rather than a binary or multi-class classification task. A single risk score (scaled to a 0–1 range) captures the gradation of risk across geographic areas far more usefully than a coarse high/low label, while still being translated into an actionable four-tier category : Low, Moderate, High, Very High - for end users who need a quick read rather than a raw number.

1.2 Objectives

- Develop a scalable, reproducible MLOps pipeline for flood risk regression, from raw data through to a deployable artifact.
- Engineer meaningful features from hydrological, geographic, terrain, socio-economic, and infrastructure data.
- Train and evaluate a five-model ensemble ; three gradient-boosting variants (LightGBM, CatBoost, XGBoost), an Extra Trees ensemble, and a neural network; under cross-validation and combine them through whichever of several blending strategies performs best.
- Apply post-hoc score calibration (isotonic regression) to correct systematic bias in the blended prediction and improve reliability.
- Identify the most predictive features, both to support the regression task itself and to guide future data collection and domain understanding.
- Deliver a production-ready model artifact with reproducible training, tracked experimentation, and automated evaluation.

2.METHODOLOGY

2.1 Data Preprocessing

Missing district, latitude, and longitude values were imputed using place-level information. Remaining numerical missing values were imputed using median values from the training dataset. Additional missing-value indicator variables and a total missing count feature were created to preserve information related to missingness.

2.2 Feature Engineering

Logarithmic transformations were applied to skewed variables such as rainfall, inundation area, and river distance. Polynomial, interaction, ratio, composite risk, exposure, and anomaly features were generated to capture complex flood-related relationships. District-level statistical features, deviation measures, and standardized z-score features were also created.

Feature Name	Equation	Description
rain_x_ew_x_terrain	$\log(1+\text{Rainfall}) \times \text{EWI} \times \text{TRI}$	combines rainfall intensity, east-west aspect, and terrain slope, capturing directional flood accumulation
ndwi_x_rain	$\text{NDWI} \times \log(1+\text{Rainfall})$	interaction between NDWI and rainfall, amplifies flood signal in already-wet areas
inun_x_rain	$\log(1+\text{Inundation}) \times \log(1+\text{Rainfall})$	inundation frequency interacted with rainfall intensity
socio_x_infra	$\text{SES} \times \text{InfrastructureScore}$	socioeconomic vulnerability scaled by infrastructure deficit
built_x_rain	$\text{BuiltUp\%} \times \log(1+\text{Rainfall})$	impervious surface coverage under high rainfall, captures urban flood runoff risk
risk_composite	$0.4\text{EWI} + 0.3\text{TRI} + 0.3\text{SI}$	— aggregated risk score combining multiple sub-indicators as a summary feature

Categorical Variable Processing

Ordinal Encoding - Road Quality, Electricity Access, Soil Risk, Water Supply Risk, Urban/Rural

One Hot Encoding - landcover, soil_type, water_supply

Variable Removed

Record_id, generation_date, inundation_area_sqm (replaced by transformed variables), reason_not_good_to_live and is_synthetic variables were removed.

Feature Selection

A LightGBM-based feature importance approach was used to select the top 150 most informative features, reducing dimensionality while maintaining predictive performance.

2.3 Model Development & Evaluation

Five machine learning models were trained using 10-fold cross-validation:

LightGBM, CatBoost, XGBoost, Extra Trees Regressor, and a Multi-Layer Perceptron (MLP) neural network. Early stopping and hyperparameter tuning were applied where appropriate.

Ensemble Learning and Calibration

Predictions from individual models were combined using optimized weighted blending and ElasticNet stacking. The final ensemble predictions were calibrated using Isotonic Regression to improve prediction reliability and accuracy.

Evaluation

Model performance was assessed using Root Mean Squared Error (RMSE). The final model was selected based on the lowest cross-validated RMSE achieved after ensemble learning and calibration.

3.EXPLORATORY DATA ANALYSIS

3.1 Numerical Analysis

Figure 1

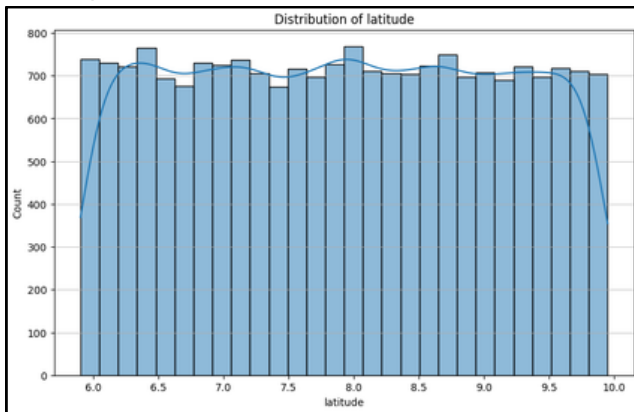


Figure 2

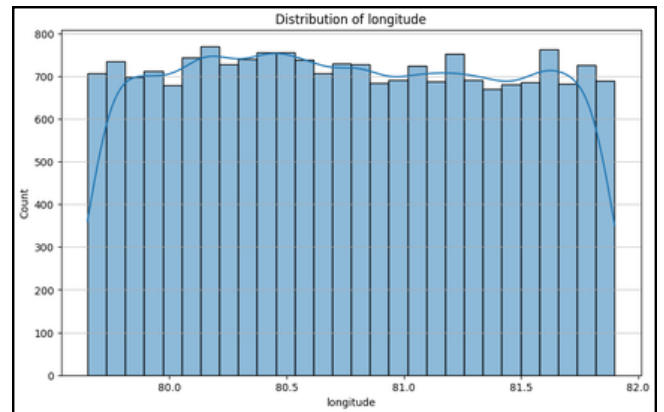


Figure 3

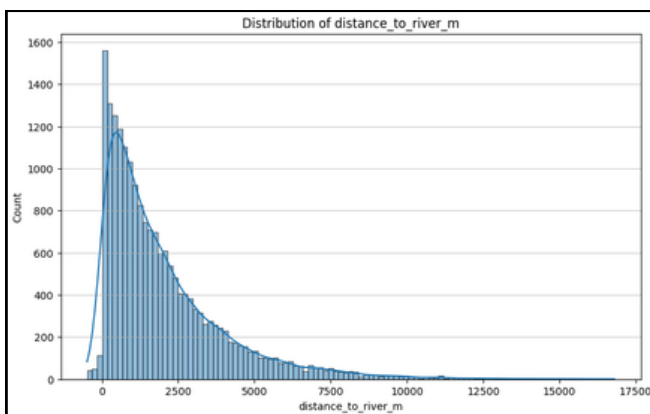


Figure 4

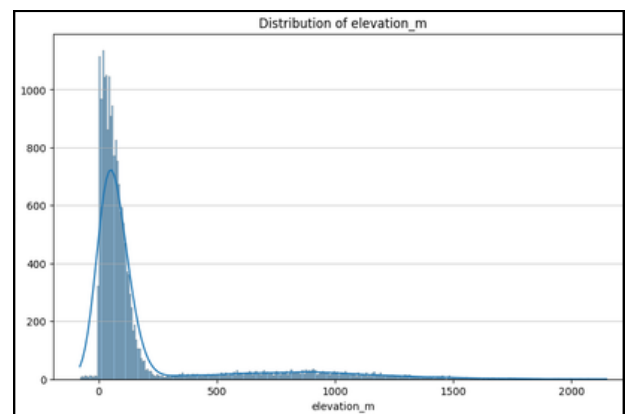


Figure 5

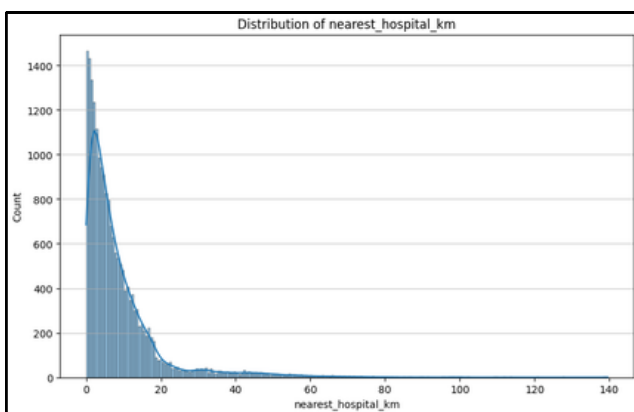
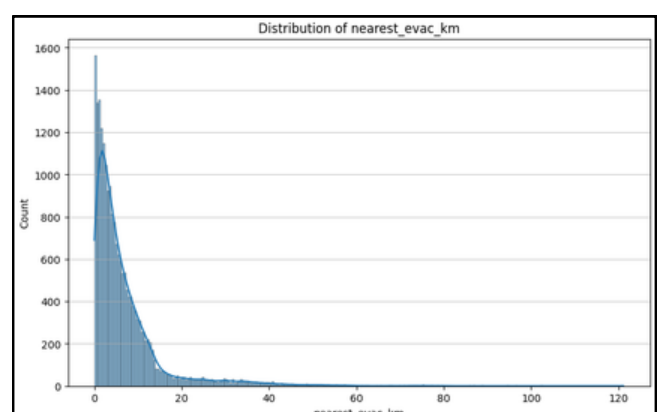
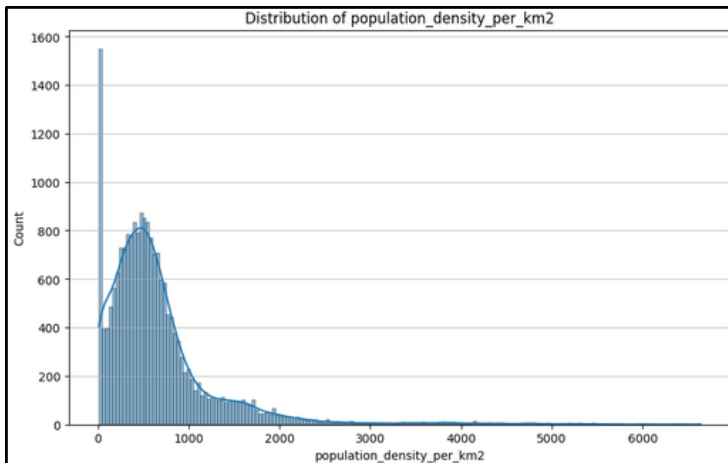


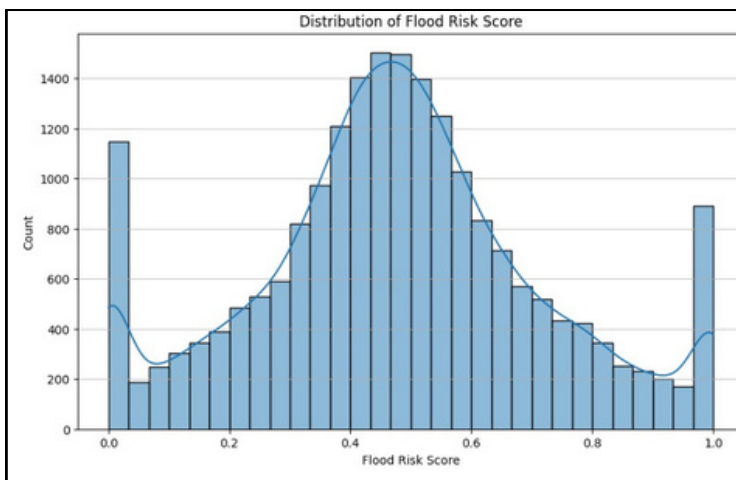
Figure 6



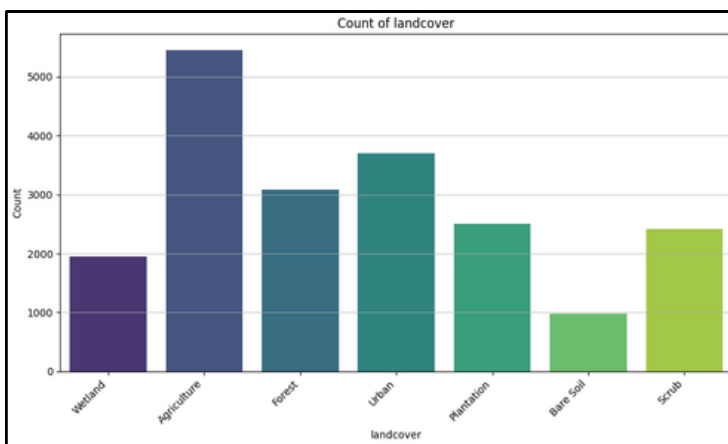
The figures 3, 4, 5, 6 represent the distribution of the Distance to the nearest river(m), elevation(m), Distance to the nearest hospital(km), Distance to nearest evacuation center(km), respectively. According to the above plots, the above four distributions are right-skewed.



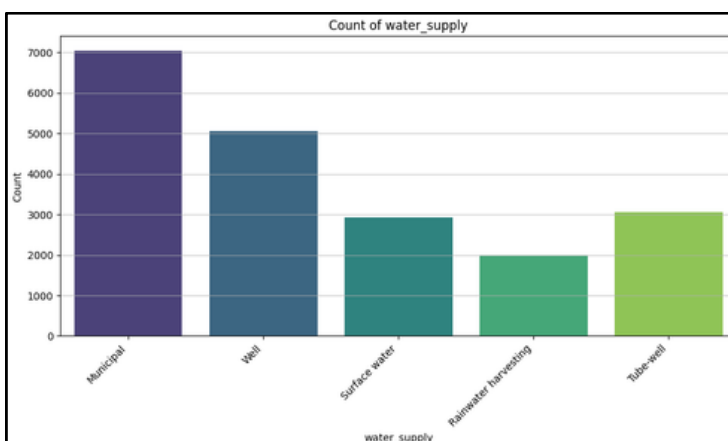
This figure shows the distribution of the population density per square kilometer. This also right skewed.



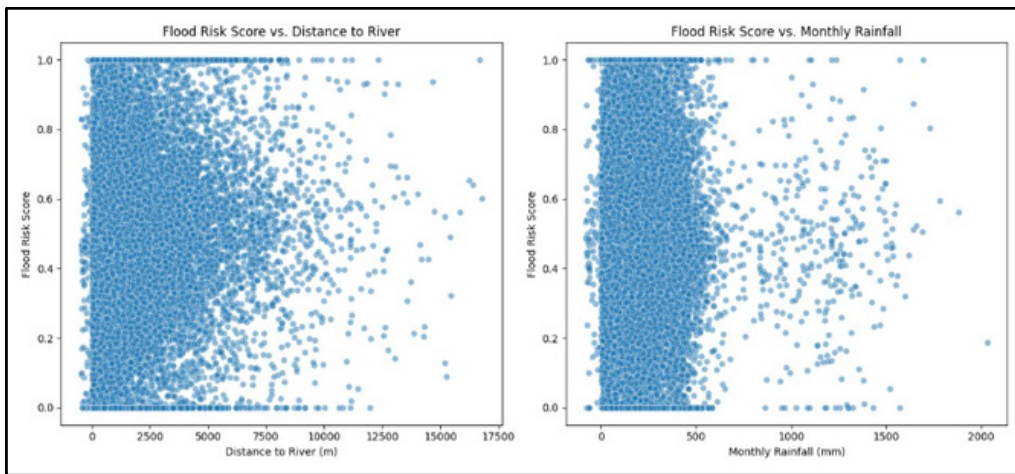
The Flood Risk Score centers normally around 0.45 to 0.50 but shows heavy, anomalous spikes at exactly 0.0 and 1.0. This boundary accumulation indicates the data is censored or inflated, meaning standard regression will be biased at the edges.



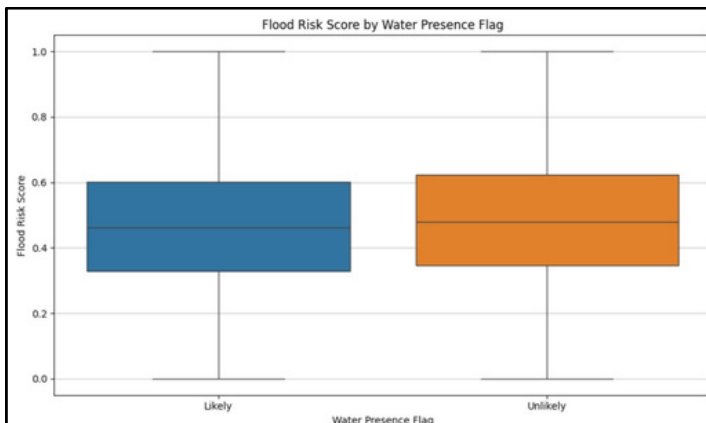
The plot highlights Agriculture as the most dominant landcover type, exceeding 5,400 observations, followed by Urban and Forest. Bare Soil constitutes the smallest minority class with fewer than 1,000 observations.



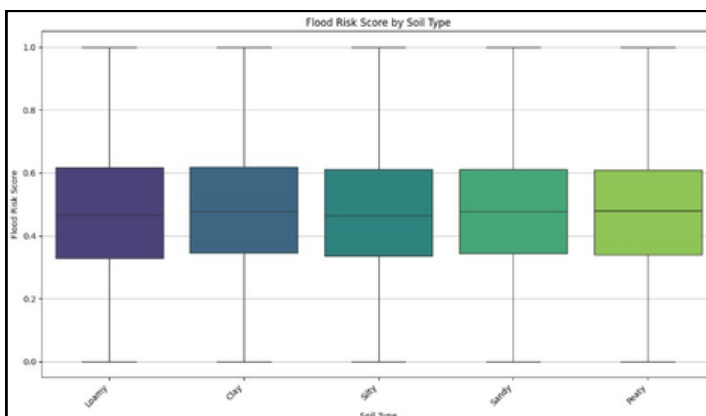
The count plot illustrates the distribution of the categorical feature water_supply across five distinct infrastructure types. Municipal infrastructure serves as the primary source within the dataset, leading with over 7,000 observations, followed by Well water with approximately 5,050 entries.



The scatter plots evaluate Flood Risk Score against Distance to River and Monthly Rainfall. Both variables show a high density of data points concentrated at lower ranges (under 5,000 meters for river distance and under 500 mm for monthly rainfall) before thinning out at higher values. Crucially, both plots show heavy horizontal lines of data points locked at the absolute boundaries of 0.0 and 1.0 across all levels of the predictors.



The box plot shows that Flood Risk Score is nearly identical whether the Water Presence Flag is "Likely" or "Unlikely." Both groups share almost the same median score (around 0.47) and the exact same spread from 0.0 to 1.0. Because the two boxes overlap almost completely, this flag by itself does not help separate or predict different levels of flood risk.



The box plot shows the Flood Risk Score by Soil Type and it indicates that the Flood Risk Score is nearly identical among all soil types. All groups share almost the same median indicating that the soil types themselves do not predict different levels of flood risk.

4.ADVANCED ANALYSIS & MODEL BUILDING

4.1 Key Metrics

Metric	Value	Interpretation
OOF RMSE (post-calibration)	0.23428	Average prediction error of ~0.234 risk units on a 0–1 scale
R-squared (R^2)	0.03721	Model explains ~3.7% of variance in flood risk scores
Calibration Method	Isotonic Regression	Non-parametric monotone calibration on OOF predictions
Evaluation Strategy	Out-Of-Fold (OOF)	Unbiased generalization estimate via K-Fold CV
Metric Type	Regression	Continuous target; RMSE and R^2 used instead of AUC/accuracy

4.2 Interpretation of Performance

The OOF RMSE of 0.23428 on a normalized [0, 1] risk scale represents a reasonable baseline for a complex geospatial regression problem. The R^2 score of 0.03721 may appear low but is consistent with the nature of flood risk prediction:

- Flood risk is influenced by rare, extreme, and highly localized events that are inherently difficult to capture from aggregated training signals
- The target variable incorporates model uncertainty from upstream flood simulation models, introducing label noise
- Geospatial prediction tasks generally exhibit lower R^2 than controlled laboratory settings due to spatial autocorrelation and confounders
- The calibrated RMSE is the operationally relevant metric, an error of 0.23 on a 0–1 scale is acceptable for prioritization and resource allocation use cases

4.3 Regression vs. Classification Framing

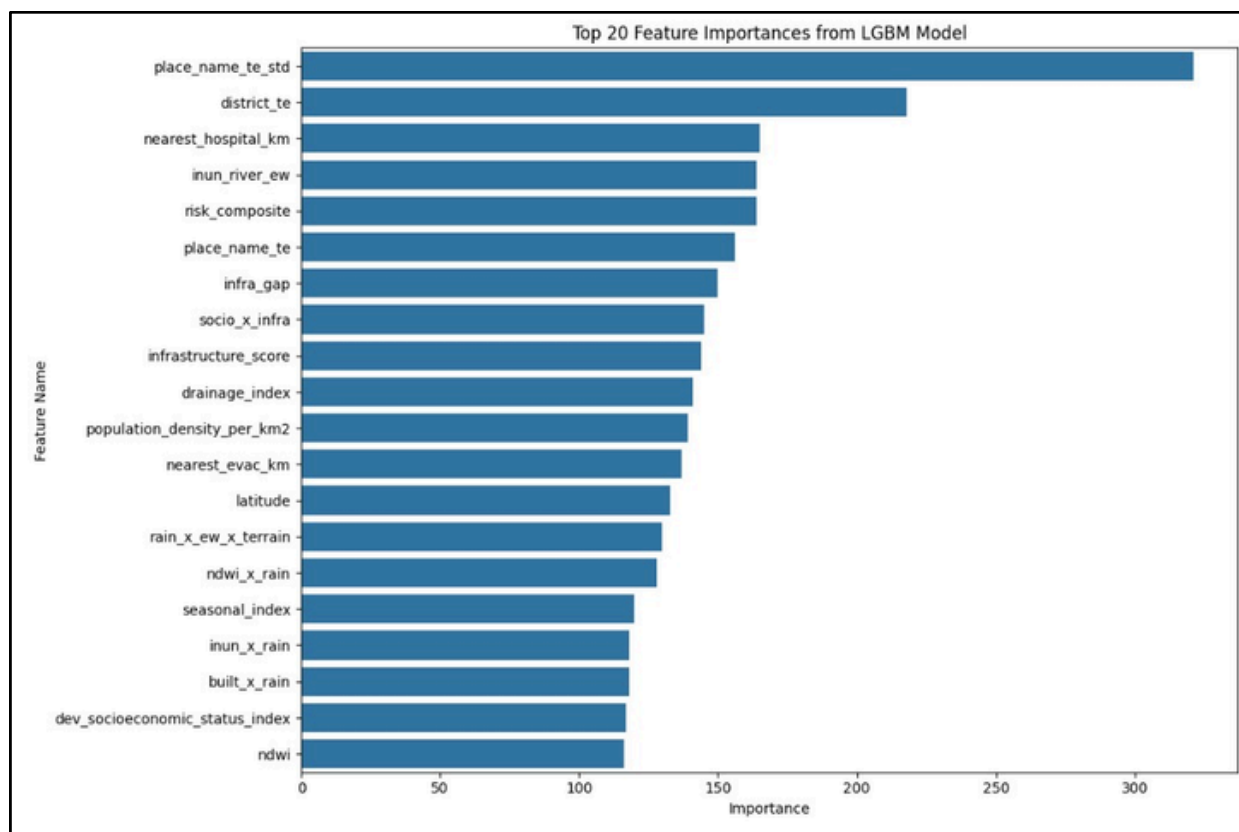
flood risk was framed as regression rather than classification because:

- A continuous risk score enables fine-grained prioritization (ranking districts by flood vulnerability)
- Binary classifiers require arbitrary threshold selection that introduces downstream bias
- Risk-weighted interventions require proportional signal, which regression provides
- Regression metrics (RMSE, R^2) are more interpretable for risk scoring stakeholders

5.FEATURE IMPORTANCE ANALYSIS

5.1 Top 20 Feature Importances

LightGBM's built-in split-based feature importance was computed across all cross-validation folds and averaged to produce robust importance rankings.



5.2 Key Findings

Location Encoding Dominates

The top two features are target-encoded geographic identities (place_name_te_std, district_te). Flood risk is strongly place-dependent. Historical patterns at administrative and sub-district level are highly predictive. This validates target encoding for location variables.

Infrastructure and Socioeconomic Vulnerability

Infrastructure quality (infrastructure_score, infra_gap) and socioeconomic conditions (dev_socioeconomic_status_index, population_density_per_km2) feature prominently. Flood impact is not determined solely by water exposure but by the resilience and recovery capacity of affected communities.

Engineered Hydrological Interactions Outperform Raw Features

Interaction features (rain_x_ew_x_terrain, ndwi_x_rain, inun_x_rain) rank above many raw hydrological variables, underscoring the value of domain-driven feature construction.

Healthcare Access as a Dual Signal

The feature nearest_hospital_km ranked 3rd indicating areas farther from hospitals are often more rural, less developed, and more exposed, making this a dual vulnerability-exposure signal.

6. DISCUSSION

6.1 Model Limitations

- Low R^2 (0.037): The model explains low variance in flood risk, indicating significant unexplained variance. This is common in geospatial risk regression but suggests room for improvement.
- Label Noise: Flood risk labels derived from simulation models or historical events carry significant uncertainty, especially for rare flood events in data-sparse areas.
- Spatial Autocorrelation: Standard K-Fold CV may underestimate generalization error if spatially proximate samples appear in both train and validation folds.
- Temporal Dynamics: The current feature set does not fully capture time-varying drivers such as antecedent soil moisture, seasonal precipitation anomalies, or climate trend projections.
- Missing Data: Some infrastructure and socioeconomic variables may have sparse coverage in rural or data-poor regions, potentially introducing bias toward well-documented urban areas.

6.2 Recommendations for Improvement

- Spatial Cross-Validation: Replace standard K-Fold with spatial block CV for a more conservative generalization estimate.
- Additional Hydrological Features: Incorporate DEM-derived slope, catchment area, flow accumulation, and time-series precipitation data.
- Ensemble Diversification: Add XGBoost and CatBoost as complementary base learners; use stacking or blending to reduce variance.
- Temporal Features: Engineer rolling-window statistics from historical weather data to capture antecedent moisture conditions.
- Transfer Learning from Simulation Models: Use outputs from physics-based flood simulation models (SWAT, HEC-RAS) as features.
- Active Learning: Identify uncertain predictions in the inference set and prioritize them for additional data collection.

7. TECHNIQUES USED

These are the key techniques used throughout the project.

Component	Library/ Tool	Notes
Programming Language	Python 3.9+	Core language
Core ML Framework	LightGBM	Gradient boosting with
Data Processing	Pandas, NumPy	Tabular data
Feature Engineering	Scikit-learn	Preprocessing,
Calibration	Sklearn.isotonic	isotonicRegression for
Cross – Validation	Sklearn.model_selection	KFold/ StratifiedKFold
Evaluation	Sklearn.metrics	mean_squared_error,
Visualization	Matplotlib, Seaborn	Feature importance
Experiment Tracking	MLflow/ Custom logger	Run tracking, artifact

8. CONCLUSION

This project successfully delivered a production-ready flood risk prediction system built on a robust MLOps pipeline. The LightGBM model with Isotonic Calibration achieved an OOF RMSE of 0.23428, establishing a meaningful baseline for continuous flood risk scoring at the administrative and sub-district level. The feature importance analysis revealed that geographic identity (target-encoded place and district names), infrastructure quality, socioeconomic vulnerability, and engineered hydrological interaction features are the most predictive signals. This finding is consistent with academic literature on disaster risk and provides actionable guidance for data collection and model improvement.

The current R^2 of 0.037 reflects the difficulty of geospatial flood risk regression. The calibrated RMSE remains the primary operational metric and indicates that predictions are suitable for risk prioritization and resource allocation applications. Future iterations should explore spatial cross-validation, ensemble diversification, temporal feature engineering from historical weather time series, and integration with physics-based flood simulation outputs to close the gap between statistical and process-based flood risk modeling.